Contents

App 1 : Login Page..... 5

App2 : Simple Calculator..... 9

App 3 : Intent……26

App 4 : Activity Life Cycle……34

App 5 : Check Box and Radio Button…….. 36

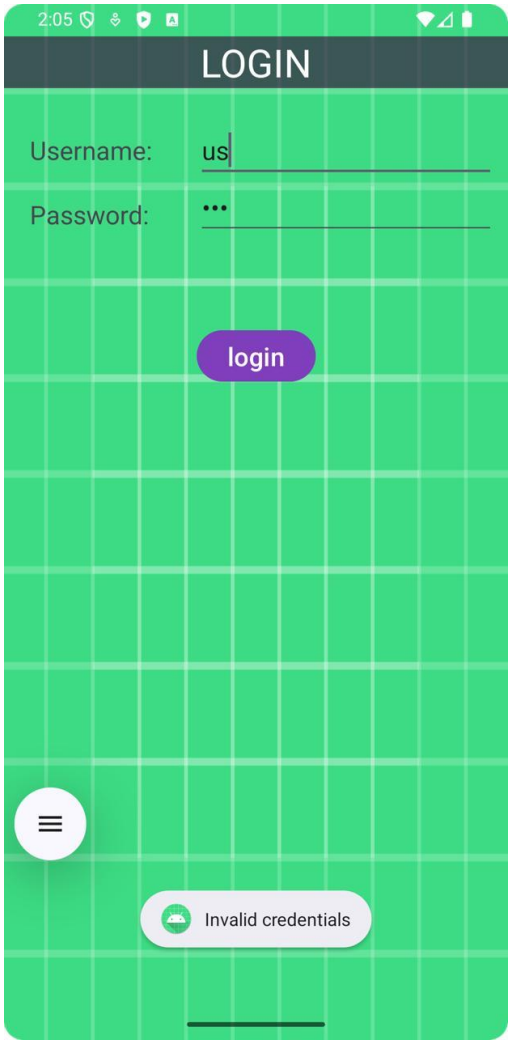
App 6 : Implicit Intent…… 42

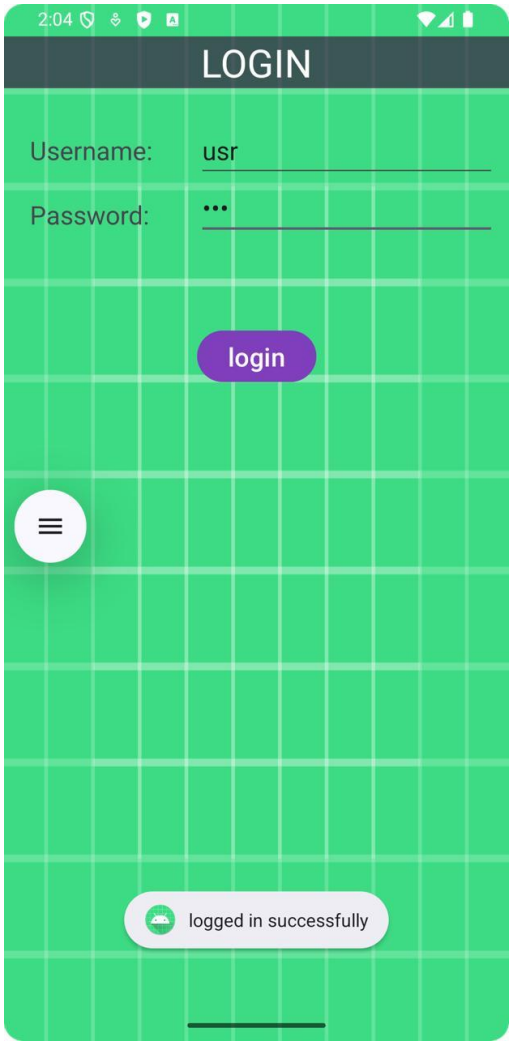
App 7 : List View…… 46

App 8 : Scroll View…….....50

App 9 : Spinner…… 54

App 10 : Shared Preferances …….. 58


2:05
LOGIN
Username:
us
Password:
login
Invalid credentials


2:04
LOGIN
Username:
usr
Password:
login
logged in successfully

Date:07/07/2026

***APP 1*: Login Page**

Aim: Design a Login page with username and password using Linear Layout and Toast Valid and Invalid credentials.

Program Code:

MainActivity.java

```
package com.example.myapplication;
import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
   Button b;
   EditText u,p;
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      b=(Button) findViewById(R.id.b);
      u=(EditText) findViewById(R.id.user);
      p=(EditText) findViewById(R.id.pass);
      b.setOnClickListener(new View.OnClickListener() {
         @Override
         public void onClick(View view) {
            String user=u.getText().toString();
            String pass=p.getText().toString();
            if (user.equals("usr")&& pass.equals("123"))
               Toast.makeText(getApplicationContext(),"logged in
successfully",Toast.LENGTH_SHORT).show();
            else
               Toast.makeText(getApplicationContext(),"Invalid
credentials",Toast.LENGTH_SHORT).show();
         }})
   }
}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   tools:context=".MainActivity"
android:background="@drawable/ic_launcher_background">
   <RelativeLayout
      android:layout_width="match_parent"
      android:layout_height="match_parent">

      <TextView
         android:id="@+id/l"
         android:layout_width="1200dp"
         android:layout_height="wrap_content"
         android:layout_alignParentEnd="true"
         android:layout_marginTop="25dp"
         android:layout_marginEnd="0dp"
         android:background="#B23B1E40"
         android:text="LOGIN"
         android:textAlignment="center"
         android:textColor="#ffffff"
         android:textSize="30dp" />

      <TextView
         android:id="@+id/u"
         android:layout_width="wrap_content"
         android:layout_height="wrap_content"
         android:text="Username:"
         android:layout_marginTop="100dp"
         android:layout_marginLeft="20dp"
         android:textSize="20dp”
         />

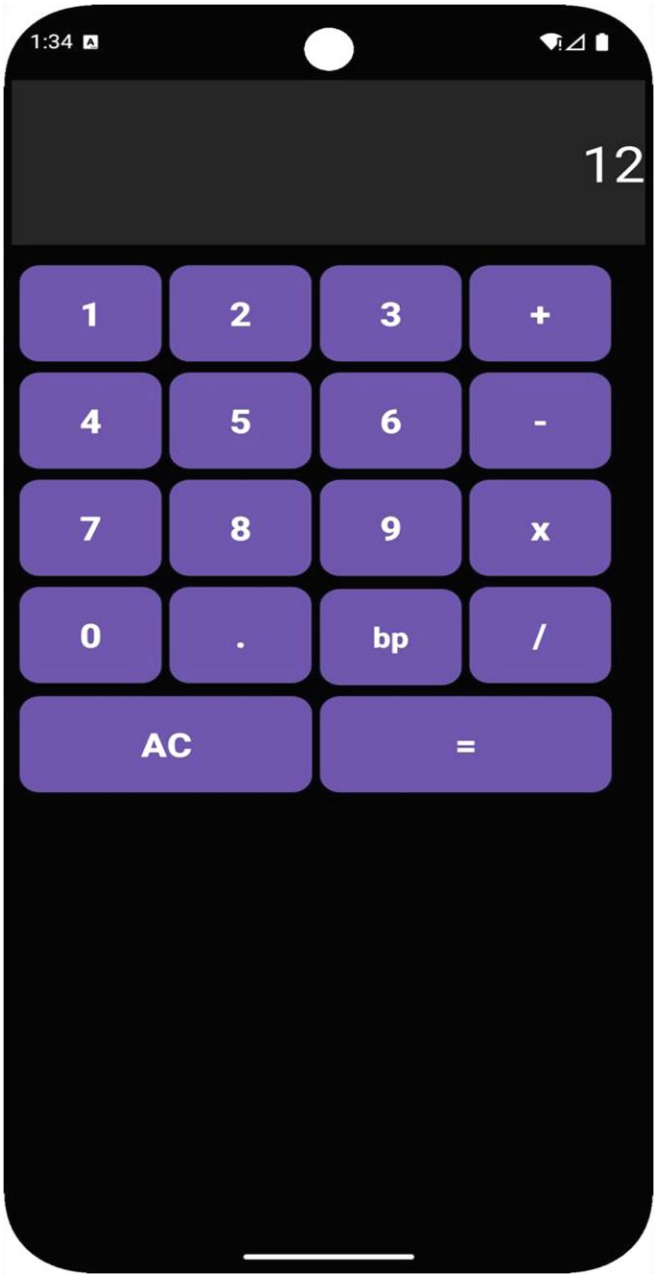
<TextView
         android:id="@+id/textView3"
         android:layout_width="wrap_content"
         android:layout_height="wrap_content"
         android:text="Password:"
         android:layout_marginTop="150dp"
         android:layout_marginLeft="20dp"
         android:textSize="20dp"
         />
<Button
         android:id="@+id/b"
         android:layout_width="wrap_content"
         android:layout_height="wrap_content"
         android:layout_marginLeft="150dp"
         android:layout_marginTop="250dp"
android:backgroundTint="#DF803BBC"
         android:text="login"
         android:textSize="20dp" />

      <EditText
         android:id="@+id/user"
         android:layout_width="wrap_content"
         android:layout_height="wrap_content"
         android:layout_marginTop="90dp"
         android:layout_marginLeft="150dp"
         android:textSize="20dp"
         android:ems="10"
         android:inputType="text"
         android:text="" />

      <EditText
         android:id="@+id/pass"
         android:layout_width="wrap_content"
         android:layout_height="wrap_content"
         android:layout_marginTop="135dp"
         android:layout_marginLeft="150dp"
         android:textSize="20dp"
         android:ems="10"
         android:inputType="textPassword" />
   </RelativeLayout>

</LinearLayout>
```

Result: The program has executed successfully and output is verified


1:34
12
1
2
3
+
4
5
6
-
7
8
9
x
0
.
bp
/
AC
=

Date: 14/07/2026

APP 2: Simple Calculator

Aim: Create a simple calculator to perform arithmetic operations

Program Code:

MainActivity.java

```
package com.example.calculator;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
public class MainActivity extends AppCompatActivity
{
   EditText t1;
   Button bac;
   String firstNumber="",operator="";
   @Override
   protected void onCreate(Bundle savedInstanceState)
   {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      t1=(EditText) findViewById(R.id.t1);
      bac=(Button) findViewById(R.id.bac);
      int b[]=new int[17];
      for(int i=0;i<=16;i++)
      {
         String s="b"+(i);
         int rid=getResources().getIdentifier(s,"id",getPackageName());
         b[i]=rid;
      }
      for(int id:b)
      {
         Button button=findViewById(id);
         if(button !=null)
         {
            button.setOnClickListener(new View.OnClickListener()
            {
               @Override
               public void onClick(View view)
```

```
            {
                handleButtonClick(view.getId());
            }});
    }}

    bac.setOnClickListener(new View.OnClickListener()
    {
        @Override
        public void onClick(View view)
        {
            t1.setText("");
        }});
}

private void handleButtonClick(int id)
{
    String txt = t1.getText().toString();
    if (id == R.id.b16)
    {
        if (!firstNumber.isEmpty() && !operator.isEmpty())
        {
            String secondNumber = txt;
            double result = 0;
            double num1 = Double.parseDouble(firstNumber);
            double num2 = Double.parseDouble(secondNumber);

            switch (operator)
            {
                case "+":
                    result = num1 + num2;
                    break;
                case "-":
                    result = num1 - num2;
                    break;
                case "x":
                    result = num1 * num2;
                    break;
                case "/":
                    result = num1 / num2;
                    break;
            }
```

```
            t1.setText(String.valueOf(result));
            firstNumber = "";
            operator = "";
        }}
      else if (id == R.id.b10)
        if (!txt.contains("."))
t1.setText(txt + ((Button) findViewById(id)).getText());
      else if (id == R.id.b12 || id == R.id.b13 || id == R.id.b14 || id == R.id.b15)
        if (!txt.isEmpty())
        {
          firstNumber = txt;
          operator = ((Button) findViewById(id)).getText().toString();
          t1.setText("");
        }
      else if (id == R.id.b11)
        if (txt.length() > 0)
          t1.setText(txt.substring(0, txt.length() - 1));
      else
        t1.setText(txt + ((Button) findViewById(id)).getText());
  }}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools
"   android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#000000"
    tools:context=".MainActivity">

    <GridLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:columnCount="4"
        android:rowCount="6"
        android:layout_margin="5dp">

        <EditText
            android:id="@+id/t1"
            android:layout_marginTop="350dp"
            android:layout_marginBottom="15dp"
            android:layout_width="match_parent"
            android:layout_height="150dp"
            android:layout_row="0"
            android:layout_column="0"
            android:ems="10"
            android:layout_columnSpan="4"
            android:inputType="text"
            android:textSize="92px"
            android:textAlignment="viewEnd"
            android:background="#20ffffff"
            android:textColor="#ffffff"
            android:text=""/>
android:background="@drawable/button_background"
            android:gravity="center"
            android:text="2"
            android:textAlignment="center"
            android:textSize="22dp"
            android:textStyle="bold"
```

```
        android:layout_marginBottom="10dp"/>

    <Button
        android:id="@+id/b3"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_row="1"
        android:layout_column="2"
        android:layout_marginLeft="5dp"
android:background="@drawable/button_back
ground"
        android:text="3"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
       android:layout_marginBottom="10dp"/>

    <Button
        android:id="@+id/b12"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="1"
        android:layout_column="3"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
     android:background="@drawable/op_bg"
        android:textColor="#ffffff"
        android:text="+"/>

    <Button
        android:id="@+id/b4"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="2"
        android:layout_column="0"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
android:background="@drawable/button_back
ground"  android:text="4"/>

    <Button
        android:id="@+id/b5"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="2"
        android:layout_column="1"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
android:background="@drawable/button_back
ground" android:text="5"/>

    <Button
        android:id="@+id/b6"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="2"
        android:layout_column="2"
android:background="@drawable/button_back
ground" android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
        android:text="6"/>

    <Button
        android:id="@+id/b13"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp
```

```
android:layout_row="2"

        android:layout_column="3"
android:background="@drawable/op_bg"
        android:textColor="#ffffff"
android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
        android:text="-"/>

    <Button
        android:id="@+id/b7"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="3"
        android:layout_column="0"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
android:background="@drawable/button_back
ground" android:text="7"/>

    <Button
        android:id="@+id/b8"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="3"
        android:layout_column="1"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
android:background="@drawable/button_back
ground" android:text="8"/>

    <Button
        android:id="@+id/b9"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="3"
        android:layout_column="2"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold”
android:layout_marginBottom="10dp"
android:background="@drawable/button_back
ground" android:text="9"/>

    <Button
        android:id="@+id/b14"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="3"
        android:layout_column="3"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
android:background="@drawable/op_bg"
        android:textColor="#ffffff"
        android:text="x"/>

    <Button
        android:id="@+id/b0"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="4"
        android:layout_column="0"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
android:background="@drawable/button_back
ground" android:text="0"/>
    <Button
```

```
        android:id="@+id/b10"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="4"
        android:layout_column="1"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
<Button
        android:id="@+id/b10"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="4"
        android:layout_column="1"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
android:background="@drawable/button_back
ground"android:text="."/>

      <Button
        android:id="@+id/b11"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="4"
        android:layout_column="2"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
android:background="@drawable/button_back
ground" android:text="bp"/>

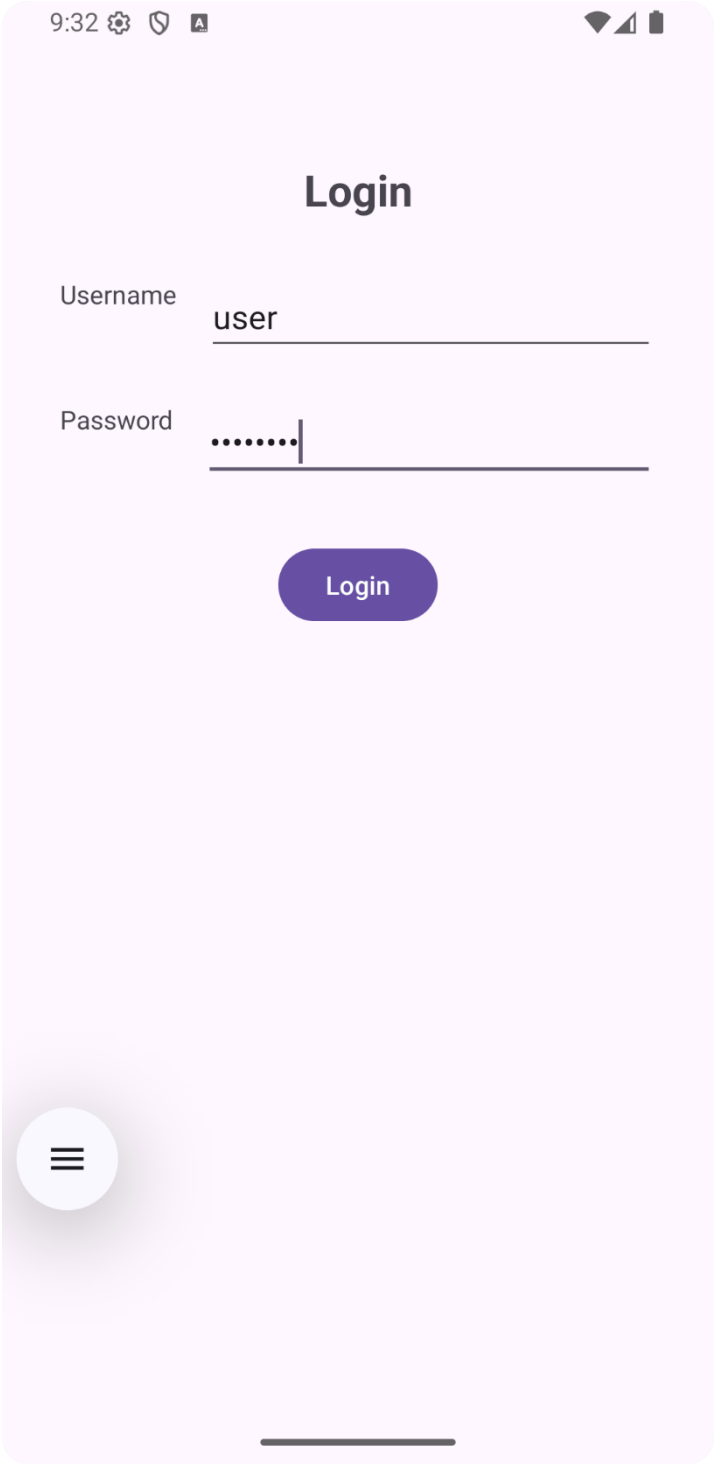
      <Button
        android:id="@+id/b15"
        android:layout_width="105dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="4"
        android:layout_column="3"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"

android:background="@drawable/op_bg"
        android:textColor="#ffffff"
        android:text="/"/>
          <Button
        android:id="@+id/bac"
        android:layout_width="215dp"
        android:layout_height="75dp"
        android:layout_row="5"
        android:layout_column="0"
        android:layout_columnSpan="2"
        android:layout_marginLeft="5dp"
        android:text="AC"
        android:textColor="#ffffff'
android:background="@drawable/op_bg"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
android:layout_marginBottom="10dp"/>

      <Button
        android:id="@+id/b16"
        android:layout_width="215dp"
        android:layout_height="75dp"
        android:layout_marginLeft="5dp"
        android:layout_row="5"
        android:layout_column="2"
        android:layout_columnSpan="2"
        android:gravity="center"
        android:textAlignment="center"
        android:textSize="28dp"
        android:textStyle="bold"
        android:layout_marginBottom="10dp"
```

```
android:background="@drawable/op_bg"
        android:textColor="#ffffff"
        android:text="="/>
    </GridLayout>
</LinearLayout>
```

Result: The program has executed successfully and output is verified


9:32
Login
Username
user
Password
Login

Date: 21/07/2026

APP 3: Intent

Aim: Create two activities , 1st activity contains login details, when clicked to submit, go to 2nd activity.

Program Code:

MainActivity.java

```
package com.example.login;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
    EditText e1,e2;
    Button b1;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
        e1=(EditText) findViewById(R.id.edittext1);
        e2=(EditText) findViewById(R.id.edittext2);
        b1=(Button) findViewById(R.id.btn1);
        b1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String username=e1.getText().toString().trim();
                String pass=e2.getText().toString();
```

```
            if(username.equals("user") && pass.equals("admin123")) {
                Toast.makeText(MainActivity.this, "Login successful", Toast.LENGTH_SHORT).show();
                Intent intent = new Intent(MainActivity.this, Homepage.class);
                startActivity(intent);
            }
            else
                Toast.makeText(MainActivity.this,"Login failed",Toast.LENGTH_SHORT).show();
        }
    });
}}
```

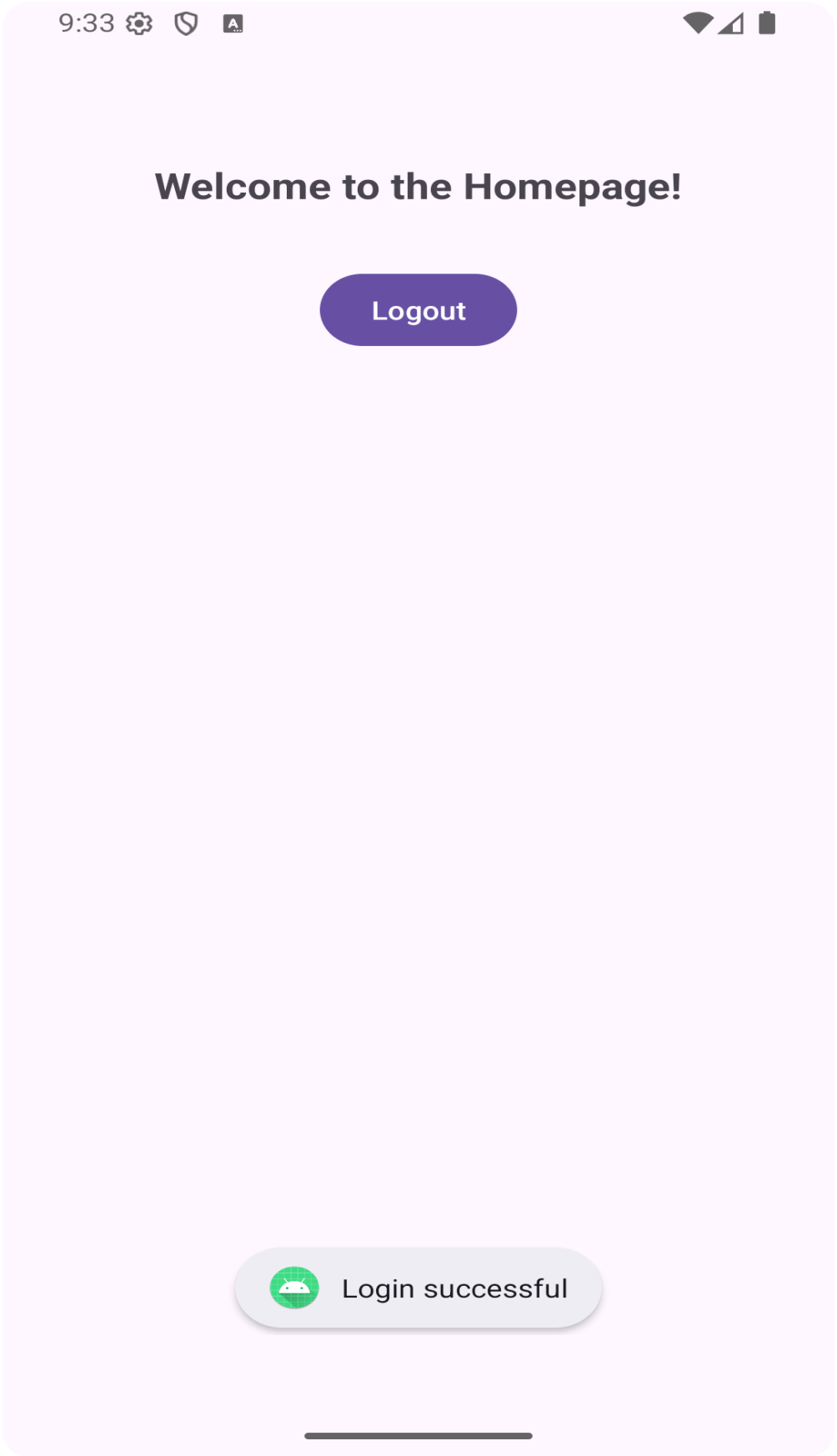
Activity_Main.xml

```
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/textView3"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Login"
        android:textSize="24sp"
        android:textStyle="bold"
        app:layout_constraintTop_toTopOf="parent"
      app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="64dp" />

    <TextView
        android:id="@+id/textView4"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Username"
app:layout_constraintTop_toBottomOf="@id/textView3"
      app:layout_constraintStart_toStartOf="parent"
        android:layout_marginTop="32dp"
        android:layout_marginStart="32dp" />
    <EditText
        android:id="@+id/edittext1"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:hint="Enter username"
        android:inputType="text"
app:layout_constraintStart_toEndOf="@id/textView4"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintTop_toTopOf="@id/textView4"
        android:layout_marginStart="16dp"
        android:layout_marginEnd="32dp" />
    <TextView
        android:id="@+id/textView5"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Password"
app:layout_constraintTop_toBottomOf="@id/edittext1"
app:layout_constraintStart_toStartOf="parent"
        android:layout_marginTop="24dp"
        android:layout_marginStart="32dp" />

    <EditText
        android:id="@+id/edittext2"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:hint="Enter password"
        android:inputType="textPassword"
```


9:33
Welcome to the Homepage!
Logout
Login successful

```
app:layout_constraintStart_toEndOf="@id/textView5"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintTop_toTopOf="@id/textView5"
        android:layout_marginStart="16dp"
        android:layout_marginEnd="32dp" />

    <Button
        android:id="@+id/btn1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Login"
app:layout_constraintTop_toBottomOf="@id/edittext2"
app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="32dp" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

Homepage.java

```
package com.example.login;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class Homepage extends AppCompatActivity {
    Button logoutButton;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_homepage);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });

        logoutButton = findViewById(R.id.logoutButton);
        logoutButton.setOnClickListener(new View.OnClickListener() {
            @Override
       public void onClick(View v) {
              // Go back to login screen
```

```
            Intent intent = new Intent(Homepage.this, MainActivity.class);
            startActivity(intent);
            finish();
        }
    });
}}
```

Activity_HomePage.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:id="@+id/main"
   android:padding="24dp">

   <TextView
      android:id="@+id/welcomeText"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Welcome to the Homepage!"
      android:textSize="20sp"
      android:textStyle="bold"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintEnd_toEndOf="parent"
      android:layout_marginTop="64dp" />
   <Button
      android:id="@+id/logoutButton"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Logout"
app:layout_constraintTop_toBottomOf="@id/welcomeText"

app:layout_constraintStart_toStartOf="parent"
app:layout_constraintEnd_toEndOf="parent"
      android:layout_marginTop="32dp" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

Result: The program has executed successfully and output is verified

Date : 28/07/2026

APP 4: Activity Life Cycle

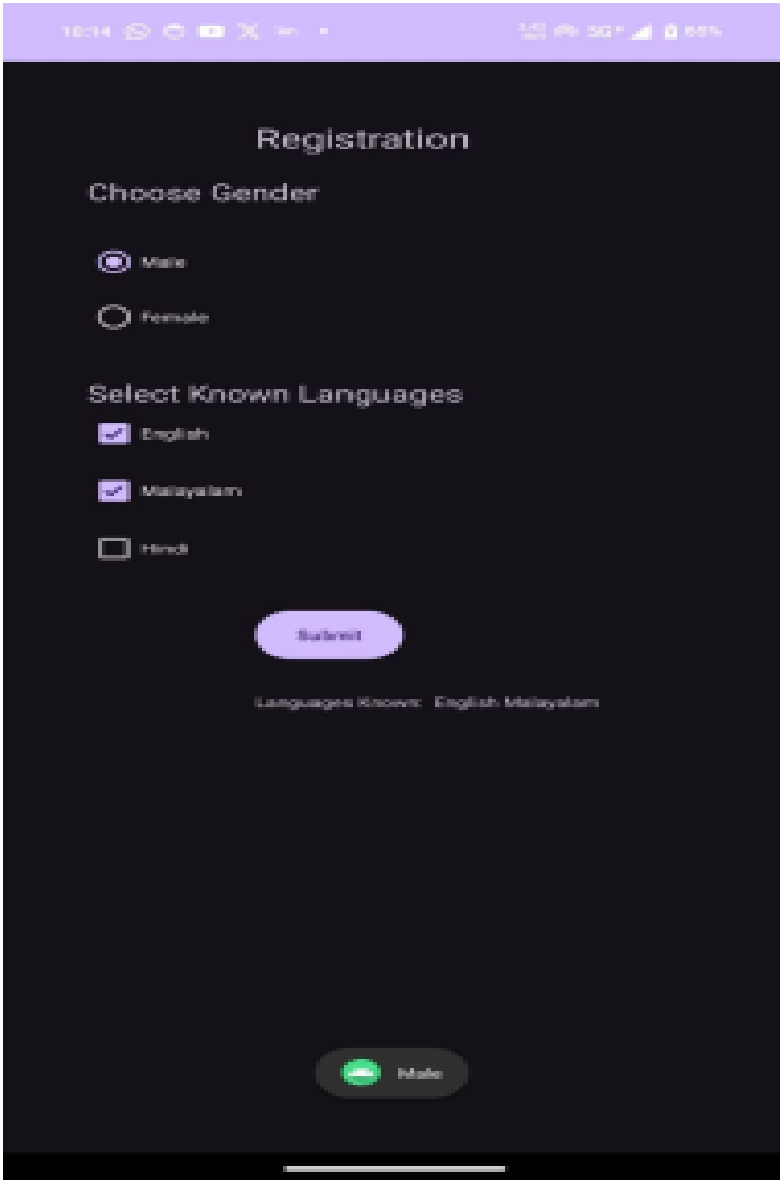
Aim: Create an app to demonstrate Activity Life Cycle.

Program Code :

MainActivity.java

```
package com.example.start;
import androidx.appcompat.app.AppCompatActivity;
import android.nfc.Tag;
import android.os.Bundle;
import android.util.Log;
public class MainActivity extends
AppCompatActivity
{
 @Override
 protected void onCreate(Bundle savedInstanceState)
 {
 super.onCreate(savedInstanceState);
 setContentView(R.layout.activity_main);
 Log.d("Appstate","Created");
 }
 @Override
 protected void onStart()
 {
 super.onStart();
 Log.d("Appstate","Starting");
 }
 @Override
 protected void onResume()
 {
 super.onResume();
 Log.d("Appstate","Resuming");
 }
 @Override
 protected void onPause()
 {
 super.onPause();
 Log.d("Appstate","Paused");
 }
 @Override
 protected void onStop()
 {
 super.onStop();
 Log.d("Appstate","Stopped");
 }
 @Override
 protected void onRestart()
 {
 super.onRestart();
 Log.d("Appstate","Restarted");
 }
 @Override
 protected void onDestroy()
 {
 super.onDestroy();
 Log.d("Appstate","Destroyed");
 }
}
```

Result: The program has executed successfully and output is verified


Registration
Choose Gender
Male
Female
Select Known Languages
English
Malayalam
Hindi
Submit
Languages Known: English Malayalam
Male

Date : 04/08/2026

APP 5: Checkbox and Radiobutton

Aim : Create an app to demonstrate the usage of Checkbox and Radio buttons

Program Code :

MainActivity.java:

```
package com.example.radiocheck;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.TextView;
import android.widget.Toast;
public class MainActivity extends AppCompatActivity
{
Button b1;
RadioGroup rg;
RadioButton r1,r2;
CheckBox e,m,h;
TextView t;
@Override
protected void onCreate(Bundle savedInstanceState)
{
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
b1=(Button) findViewById(R.id.submit);
r1=(RadioButton) findViewById(R.id.male);
r2=(RadioButton) findViewById(R.id.female);
e=(CheckBox)findViewById(R.id.e);
m=(CheckBox)findViewById(R.id.m);
h=(CheckBox)findViewById(R.id.h);
t=(TextView)findViewById(R.id.tv);
b1.setOnClickListener(new View.OnClickListener()
{
@Override
public void onClick(View view)
{
String s[]=new String[3];
t.setText("Languages Known: ");
if(e.isChecked())
t.setText(t.getText().toString()+" "+e.getText());
```

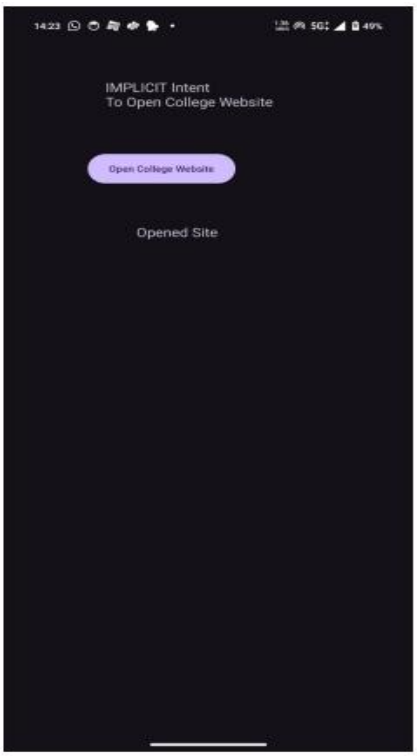
```
if(h.isChecked())
t.setText(t.getText().toString()+" "+h.getText());
}
});
rg=(RadioGroup) findViewById(R.id.rg);
rg.setOnCheckedChangeListener(new RadioGroup.OnCheckedChangeListener()
{
@Override
public void onCheckedChanged(RadioGroup group, int checkedId)
{
if(checkedId==R.id.female)
{ Toast.makeText(getApplicationContext(),r2.getText().toString(),Toast.LENGTH_SHORT).show();
}
else
{ Toast.makeText(getApplicationContext(),r1.getText().toString(),Toast.LENGTH_SHORT).show();
}
}
});
}
}
```

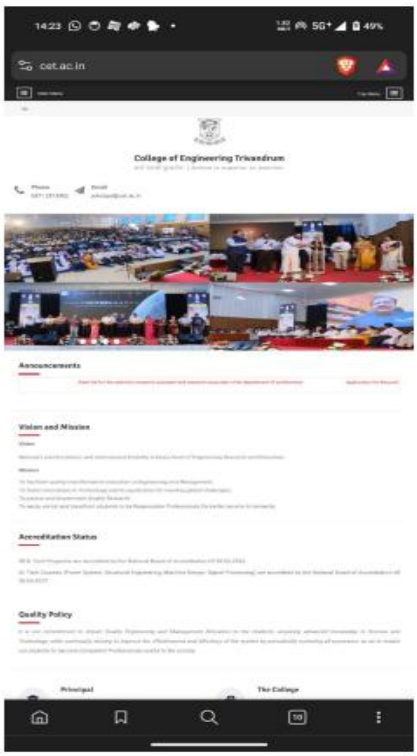
Activity_home_page.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tool"
 android:layout_width="match_parent"
 android:layout_height="match_parent"
 tools:context=".MainActivity">
 <RelativeLayout
 android:layout_width="match_parent"
 android:layout_height="match_parent">
 <TextView
 android:layout_width="wrap_content"
 android:layout_height="30dp"
 android:layout_marginTop="50dp"
 android:layout_marginLeft="150dp"
 android:id="@+id/h1"
 android:text="Registration"
 android:textSize="24dp"/>
 <TextView
 android:layout_width="wrap_content"
 android:layout_height="30dp"
 android:layout_marginTop="100dp"
 android:layout_marginLeft="50dp"
android:id="@+id/gender"
 android:text="Choose Gender"
 android:textSize="20dp"/>
 <RadioGroup
 android:layout_width="250dp"
 android:layout_height="wrap_content"
 android:id="@+id/rg"
 android:layout_marginTop="150dp"
 android:layout_marginLeft="50dp">
 <RadioButton
 android:layout_width=
"wrap_content"
 android:layout_height=
"wrap_content"
 android:text="Male"
 android:id="@+id/male"/>
 <RadioButton
 android:layout_width=
"wrap_content"
 android:layout_height=
"wrap_content"
```

```
android:text="Female"
 android:id="@+id/female"/>
 </RadioGroup>
 <TextView
 android:layout_width="wrap_content"
android:layout_height="30dp"
 android:layout_marginTop="275dp"
 android:layout_marginLeft="50dp"
 android:id="@+id/lang"
 android:text="Select Known Languages"
 android:textSize="20dp"/>
 <CheckBox
 android:id="@+id/e"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:text="English"
 android:layout_marginTop="300dp"
 android:layout_marginLeft="50dp"/>
 <CheckBox
 android:id="@+id/m"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:text="Malayalam"
 android:layout_marginTop="350dp"
 android:layout_marginLeft="50dp"/>
 <CheckBox
 android:id="@+id/h"
android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:text="Hindi"
 android:layout_marginTop="400dp"
 android:layout_marginLeft="50dp"/>
 <Button
 android:layout_width="wrap_content"
 android:layout_height="50dp"
 android:layout_marginTop="475dp"
 android:layout_marginLeft="150dp"
 android:text="Submit"
 android:id="@+id/submit"/>
 <TextView
 android:layout_width="wrap_content"
 android:layout_height="50dp"
 android:layout_marginTop="550dp"
 android:layout_marginLeft="150dp"
 android:text=""
 android:id="@+id/tv"/>
 </RelativeLayout>
</LinearLayout>
```

Result: The program has executed successfully and output is verified.


IMPLICIT Intent
To Open College Website
Open College Website
Opened Site


cet.ac.in
College of Engineering Trivandrum
Announcements
Vision and Mission
Accreditation Status
Quality Policy

Date : 11/08/2026

APP 6: Implicit Intent

Aim : Create an app to demonstrate the usage of implicit intent by opening a website form the app

Program Code :

MainActivity.java:

```
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity
{
 Button b;
 TextView t;
 @Override
 protected void onCreate(Bundle savedInstanceState)
 {
 super.onCreate(savedInstanceState);
 EdgeToEdge.enable(this);
 setContentView(R.layout.activity_main);
 b=(Button) findViewById(R.id.b1);
 t=(TextView) findViewById(R.id.t1);
 b.setOnClickListener(new View.OnClickListener()
 {
 @Override
 public void onClick(View view) {
 Intent intent = new Intent(Intent.ACTION_VIEW);
 intent.setData(Uri.parse("https://www.cet.ac.in"));
 startActivity(intent);
 t.setText("Youtube Opened");
 }
 });
 }
}
```

Activity_home_page.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools" android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">

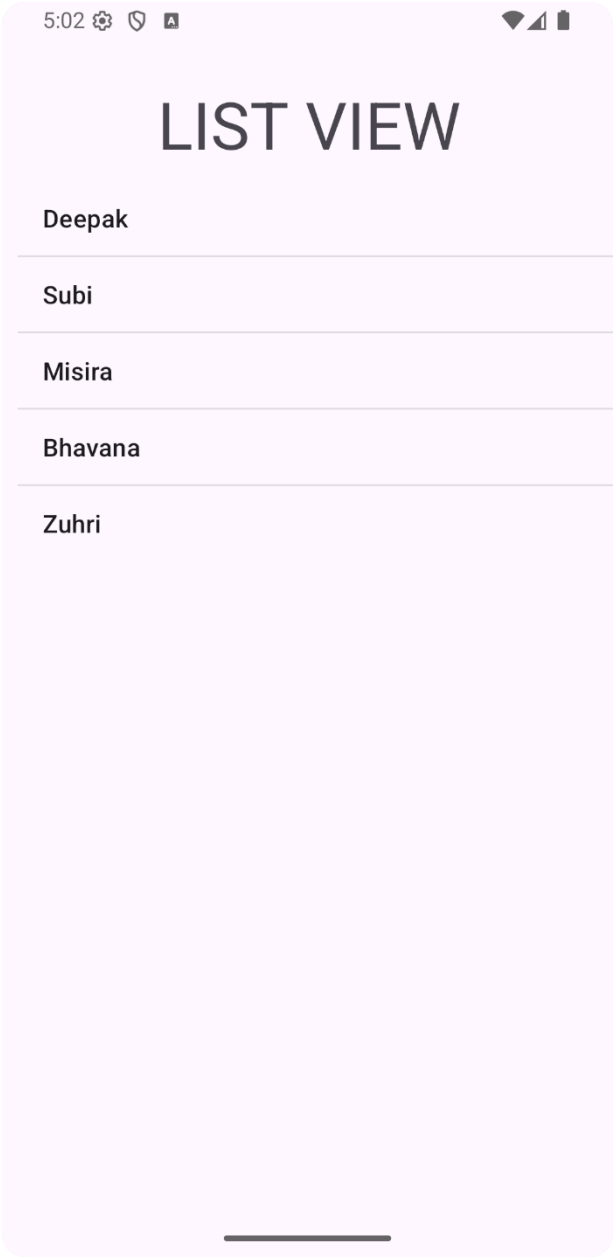
<RelativeLayout
 android:layout_width="wrap_content"
 android:layout_height="wrap_content">

 <TextView
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:text="IMPLICIT Intent \nTo
Open College Website"
 android:layout_marginLeft="115dp"
 android:layout_marginTop="100dp"
 android:textSize="20sp">
 </TextView>

 <Button
 android:id="@+id/b1"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:layout_marginLeft="95dp"
 android:layout_marginTop="200dp"
 android:text="Open College Website">
</Button>

 <TextView
 android:id="@+id/t1"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:text=""
 android:layout_marginTop="300dp"
 android:layout_marginLeft="150dp"
 android:textSize="20sp"></TextView>
 </RelativeLayout>
</LinearLayout>
```

Result: The program has executed successfully and output is verified


5:02
LIST VIEW
Deepak
Subi
Misira
Bhavana
Zuhri

Date : 18/08/2026

APP 7: List View

Aim : Create an app to create a List view with few items in it

Program Code :

MainActivity.java:

```
package com.example.listview;
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity
{
   ListView lv;
   String arr[]={"Deepak","Subi","Misira","Bhavana","Zuhri"};
   @Override
   protected void onCreate(Bundle savedInstanceState)
   {
      super.onCreate(savedInstanceState);
      EdgeToEdge.enable(this);
      setContentView(R.layout.activity_main);
      lv=(ListView) findViewById(R.id.lv);
      ArrayAdapter adapter=new ArrayAdapter<String>(this,
android.R.layout.simple_list_item_1,arr);
      lv.setAdapter(adapter);
      lv.setOnItemClickListener(new AdapterView.OnItemClickListener()
      {
         @Override
         public void onItemClick(AdapterView<?> parent, View view, int position, long id)
         {
            Toast.makeText(MainActivity.this,"Name Clicked: "+adapter.getItem(position),
                  Toast.LENGTH_SHORT).show();
         }
      });
   }
```

Activity_home_page.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
 android:id="@+id/main"
 android:layout_width="match_parent"
 android:layout_height="match_parent"
 tools:context=".MainActivity">

<RelativeLayout
 android:layout_width="wrap_content"
 android:layout_height="wrap_content">

 <TextView
android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:layout_marginTop="50dp"
 android:layout_marginLeft="100dp"
 android:text="LIST VIEW"
 android:textSize="42dp">
</TextView>

 <ListView
 android:id="@+id/lv"
 android:layout_width="wrap_content"
 android:layout_height="wrap_content"
 android:layout_marginTop="115dp"
 android:layout_marginBottom="50dp"
 android:layout_marginLeft="10dp">
</ListView>

</RelativeLayout>
</LinearLayout>
```

Result: The program has executed successfully and output is verified.


13:24
0.03 KB/S
5G+
57%
TextView 1
TextView 2
TextView 3
TextView 4
TextView 5
TextView 6
TextView 7
TextView 8
TextView 9
TextView 10
TextView 11
TextView 12
TextView 13
TextView 14
TextView 15

Date : 08/09/2026

APP 8 : Scroll View

Aim : Create an app to create a spinner and select an item from it.

Program Code :

MainActivity.java:

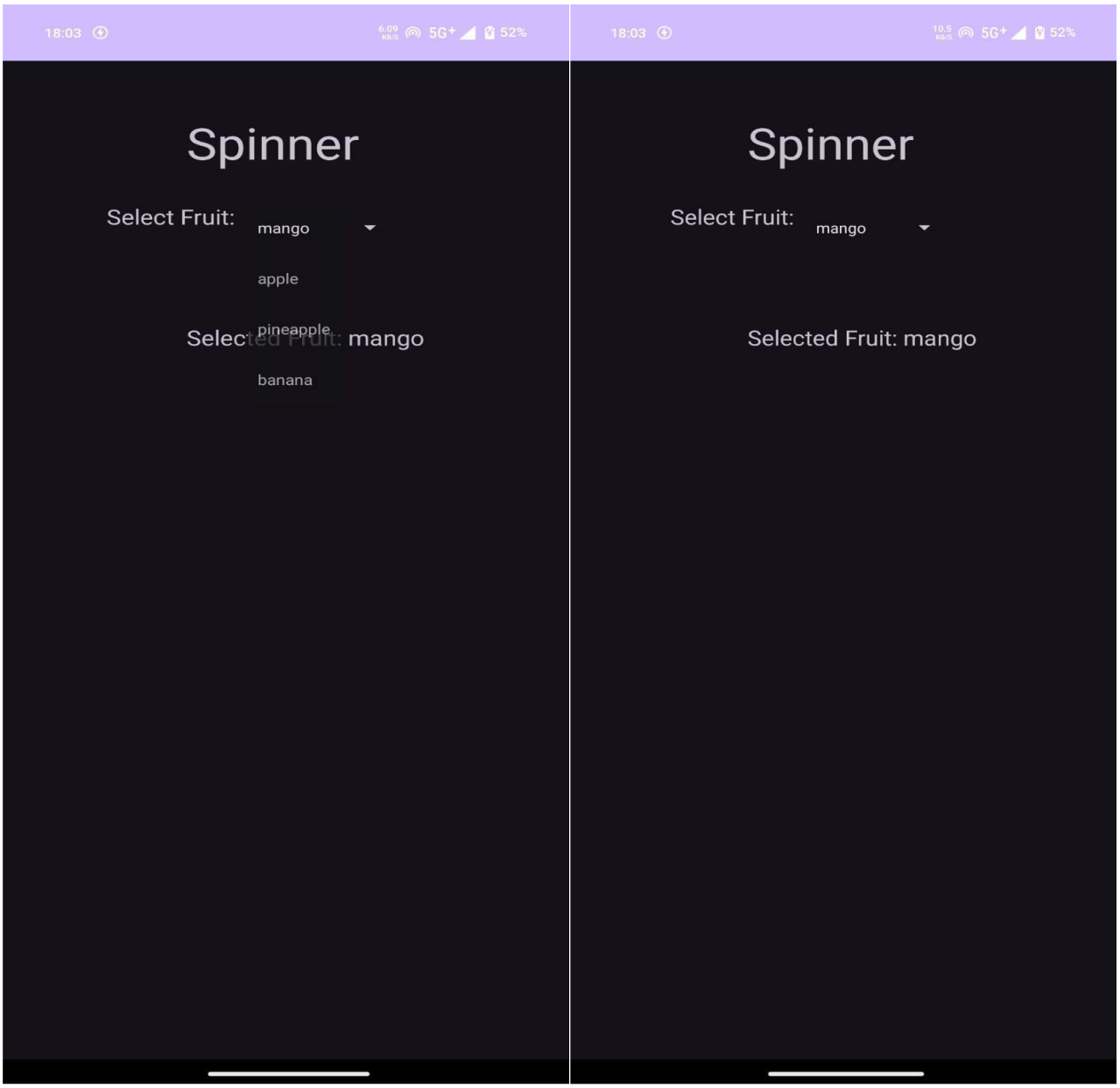
```
package com.example.scrollview;
import android.os.Bundle;
import android.widget.LinearLayout;
import android.widget.TextView;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity
{
 @Override
 protected void onCreate(Bundle savedInstanceState)
 {
 super.onCreate(savedInstanceState);
 EdgeToEdge.enable(this);
 setContentView(R.layout.activity_main);
 LinearLayout linearLayout = findViewById(R.id.linearLayoutContainer);
 for (int i = 1; i <= 20; i++)
 {
 TextView textView = new TextView(this);
 textView.setText("TextView " + i);
 textView.setTextSize(45);
 textView.setPadding(16, 16, 16, 16);
 linearLayout.addView(textView);
 }}}
```

Activity_home_page.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
 android:id="@+id/main"
 android:layout_width="match_parent"
 android:layout_height="match_parent
 android:padding="16dp"
 tools:context=".MainActivity">
 <LinearLayout
 android:id="@+id/linearLayoutContainer"
 android:layout_width="match_parent"
 android:layout_height="wrap_content"
 android:orientation="vertical">
```

</LinearLayout> | </ScrollView>

Result: The program has executed successfully and output is verified


18:03
Spinner
Select Fruit:
mango
apple
pineapple
banana
Selected Fruit: mango
18:03
Spinner
Select Fruit:
mango
Selected Fruit: mango

Date : 15/09/2026

APP 9 : Spinner

Aim : Create an app to create a spinner and select an item from it.

Program Code :

MainActivity.java:

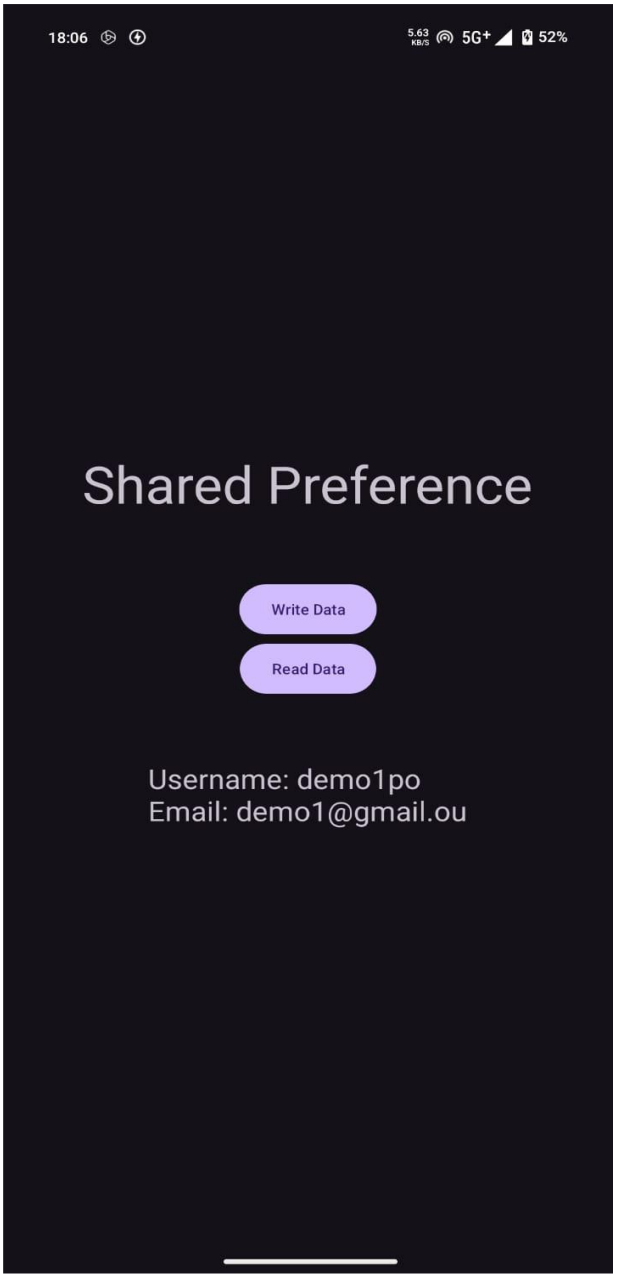
```
package com.example.spinner;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.Spinner;
import android.widget.TextView;
import android.widget.Toast;
import java.util.ArrayList;
public class MainActivity extends AppCompatActivity
{
Spinner s;
TextView t1;
@Override
protected void onCreate(Bundle savedInstanceState)
{
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
s=(Spinner)findViewById(R.id.sp);
t1=(TextView) findViewById(R.id.t1);
ArrayList <String> al=new ArrayList<String>();
al.add("mango");
al.add("apple");
al.add("pineapple");
al.add("banana");
ArrayAdapter ad=new ArrayAdapter<>(this, androidx.appcompat.R.layout.support_simple_
spinner_dropdown_item,al);
s.setAdapter(ad);
s.setOnItemSelectedListener(new AdapterView.OnItemSelectedListener()
{
@Override
public void onItemSelected(AdapterView<?> parent, View view, int position, long id)
{
t1.setText("Selected Fruit: "+al.get(position));
}
```

```
@Override
public void onNothingSelected(AdapterView<?> parent)
{
t1.setText("Select a Fruit !!!!");
}
});
}
}
```

Activity_home_page.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">
<RelativeLayout
android:layout_width="wrap_content"
android:layout_height="wrap_content">
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="150dp"
android:layout_marginTop="50dp"
android:text="Spinner"
android:textSize="42dp"></TextView>
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="85dp"
android:layout_marginTop="135dp"
android:text="Select Fruit:"
android:textSize="20dp">
</TextView>
<Spinner
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="200dp"
android:layout_marginTop="135dp"
android:id="@+id/sp"></Spinner>
<TextView
android:id="@+id/t1"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginLeft="150dp"
android:layout_marginTop="250dp"
android:text=""
android:textSize="20dp"></TextView>
</RelativeLayout>
</LinearLayout>
```

***Result*:** The program has executed successfully and output is verified.


18:06
5.63 KB/S 5G+ 52%
Shared Preference
Write Data
Read Data
Username: demo1po
Email: demo1@gmail.ou


18:06
25.0 KB/S 5G+ 52%
Shared Preference
Write Data
Read Data
Written Data

Date : 22/09/2026

APP 10 : Shared Preferances

Aim : Create an app to demonstrate the usage of shared preferences by writing and reading data.

Program Code :

MainActivity.java

```
import android.content.SharedPreferences;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity
{
TextView txt;
Button br,bw;
@Override
protected void onCreate(Bundle savedInstanceState)
{
super.onCreate(savedInstanceState);
EdgeToEdge.enable(this);
setContentView(R.layout.activity_main);
txt=(TextView) findViewById(R.id.text_status);
bw=(Button) findViewById(R.id.btn_write);
br=(Button) findViewById(R.id.btn_read);
String usernamedemo="demo1po";
String emaildemo="demo1@gmail.ou";
SharedPreferences sp =getSharedPreferences("sp_abc",MODE_PRIVATE);
bw.setOnClickListener(new View.OnClickListener()
{
@Override
public void onClick(View view)
{
SharedPreferences.Editor se = sp.edit();
se.putString("username",usernamedemo);
se.putString("email",emaildemo);
se.apply();
txt.setText("Written Data");
}
});
```

```
br.setOnClickListener(new View.OnClickListener()
{
@Override
public void onClick(View view) {
String u,e;
u=sp.getString("username","");
e=sp.getString("email","");
txt.setText("Username: "+u+" \nEmail: "+e);
}
});
}
}
```

Activity_home_page.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity"
android:orientation="vertical"
android:gravity="center">
<TextView
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Shared Preference"
android:textSize="42dp"/>
<Button
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Write Data"
android:layout_marginTop="50dp"
android:id="@+id/btn_write"/>
<Button
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="Read Data"
android:id="@+id/btn_read"/>
<TextView
android:id="@+id/text_status"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="50dp"
android:text="STATUS"
android:textSize="22dp"/>

</LinearLayout>
```

***Result*:** The program has executed successfully and output is verified.